DATABASES 1 – PROFF. S. CERI, S. COMAI, L. TANCA – A. Y. 2023/24

EXAM OF JUNE 6TH 2024 – DURATION: 2H

Please write down the solutions of the two parts of the exam on TWO DIFFERENT SHEETS

Part 1: QUERY LANGUAGES

The following DB schema describes the Formula 1 car races. Note that ChequeredFlag is True if the pilot completes the grand prix and that TyreType includes various predefined types (Red, Yellow, White, ...).

PILOT (PilotName, TeamName)

GRAND PRIX (GrandPrixName, Country, Continent)

RANKINGS (GrandPrixName, PilotName, QualifyingPosition, RacePosition, Points, ChequeredFlag, Total-NrOfLaps)

USEDTYRES (GrandPrixName, PilotName, TyreType, LapsCount)

Exercise 1. SQL (13 points)

A. Define the schema of the RANKINGS table, including referential integrity constraints and attribute nullability. Express the constraint that pilots who did not finish the race must always be assigned 0 points. (2 pts)

B. Extract the pilots who finished a Grand Prix using three different types of tyres. (3 pts)

C. Extract the pilots who finished in the top three in all the Grand Prix held in Europe. (4 pts)

D. Build the Constructors' Standings: write a view that returns the name of the team followed by the sum of the points obtained by its pilots in descending order of points. Then find the difference in points between the first and the last team in the ranking. (4 pts)

Exercise 2 – Formal Languages (6 pts)

Express in Relational Calculus and Datalog the following query:

E. Extract the pilots who have never finished in the top ten in European Grands Prix but arrived at least once in the top three in qualifying (considering only European Grands Prix).

Part 2: DB DESIGN AND THEORY

Exercise 3 - Theory (1pt)

Which are the SQL policies in response to a breach of a referential integrity constraint?

Exercis 4. Conceptual and Logical Design (13 pts)

- *The project evaluation is based on its completeness, correctness, legibility, minimality and clarity.*
- *Remember to specify the **identifiers** of all the entities and the **cardinality** of all the associations.*

Politecnico must manage the information about the Festival of Engineering, an event that lasts only one day. The festival involves as speakers the professors of Politecnico, each characterized by a name, a department, a discipline and a short curriculum. Each presentation includes one or more speakers, has a title, a duration, a start time and an abstract, and takes place in a classroom of Politecnico. Classrooms have a code and maximum capacity. Some presentations are replicated at different times of the day. The participants in a presentation enroll providing their first name, surname, age and educational qualification, and the enrollment is free. Each participant can attend several presentations.

The Festival also includes guided tours to the laboratories, each one characterized by its name and by the department that manages it; each guided tour is related to a laboratory, has a start time, an accompanying professor, and has a maximum number of participants, who must register for the visit. In addition, each specific classroom hosts, in different turns, different experimental activities, which have a name, a responsible professor and resource needs (for example, computers and screens).